

PHISHING EMAIL **DETECTION AND** **AWARENESS REPORT**

FUTURE INTERNS CYBER SECURITY INTERNSHIP
TASK 2- Phishing Email Detection And Awareness System

Prepared By:

Ann Maria Sony

Internship Domain:

Cyber Security

Repository:

FUTURE_CS_02

INTRODUCTION

Phishing is a cyberattack in which attackers send fraudulent emails pretending to be trusted organizations to trick users into revealing sensitive information such as usernames, passwords, banking details, or personal information. These emails often contain fake links, malicious attachments, or urgent messages designed to make users act without verifying the source.

The purpose of this report is to analyze phishing email samples, identify common phishing indicators, classify email risks, and provide practical security awareness guidelines to help users recognize and avoid phishing attacks.

OBJECTIVE

The objectives of this report are to:

- Analyze phishing email samples.
- Identify common phishing indicators.
- Classify emails as Safe, Suspicious, or Phishing.
- Explain phishing techniques in simple, non-technical language.
- Provide prevention and awareness guidelines for users.

TOOLS USED

The following tools and resources were used to perform phishing email analysis and prepare this report:

TOOL	PURPOSE
Public Phishing Email Samples	Used to analyse phishing and legitimate email examples.
Email Header Analyzer	Used to examine email headers and verify sender information
Web Browser	Used to inspect suspicious URLs email content safely
Google Docs	Used to prepare and format the report
GitHub	Used to document and submit the project

METHODOLOGY

The phishing email samples were carefully analyzed by examining the sender's email address, subject line, message content, hyperlinks, attachments, and writing style. Each email was then classified as **Safe**, **Suspicious**, or **Phishing** based on the identified phishing indicators. Finally, security awareness recommendations were prepared to help users recognize and avoid phishing attacks.

EMAIL SAMPLE 1 ANALYSIS



Dear User,

All Hotmail customers have been upgraded to Outlook.com. Your Hotmail Account services ha

Due to our new system upgrade to Outlook. In order for it to remain active
follow the link Sign in Re-activate your account to Outlook. <https://account.live.com>

Thanks,

The Microsoft account team

FIELD	DETAILS
EMAIL TYPE	Phishing Email
ORGANIZATION IMPERSONATED	Microsoft Outlook
TARGET	Hotmail/outlook users
CLASSIFICATION	Phishing
RISK LEVEL	High

PHISHING INDICATORS

- Uses a generic greeting ("Dear User") instead of the recipient's name.
- Creates a sense of urgency by asking the user to reactivate the account immediately.
- Requests the user to click a link to verify or reactivate the account.
- Attempts to impersonate a **trusted company (Microsoft Outlook)**.
- The user should always verify such requests through the **official Microsoft website**, not by clicking links in emails.

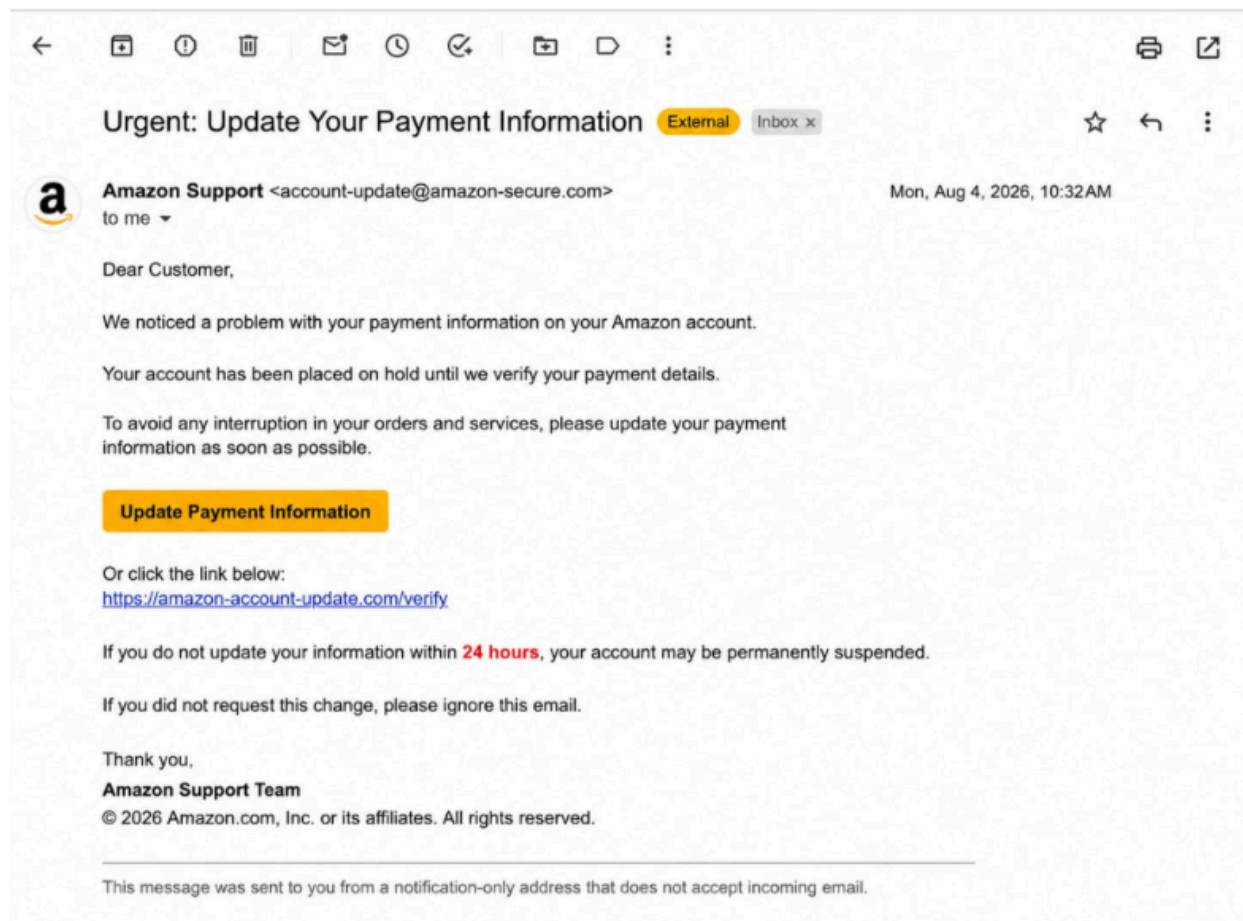
RECOMMENDED ACTIONS

- Do not click on any links or download attachments from the email.
- Do not provide personal information, passwords, or banking details.
- Verify the request by visiting the official Microsoft Outlook website directly instead of using the email link.
- Report the email as phishing to your email provider or organization's IT/security team.
- Delete the email if it is confirmed to be fraudulent.
- Enable Multi-Factor Authentication (MFA) to improve account security.

CONCLUSION

This email is classified as a phishing email because it uses social engineering techniques such as urgency, impersonation of a trusted organization, and a request to click a link. Users should always verify unexpected emails through official channels before taking any action.

EMAIL SAMPLE 2 ANALYSIS



FIELD	DETAILS
EMAIL TYPE	Phishing Email
ORGANIZATION IMPERSONATED	Amazon
TARGET	Amazon Customers
RISK LEVEL	High
CLASSIFICATION	Phishing

PHISHING INDICATORS

- Uses a **generic greeting** ("Dear Customer") instead of the recipient's actual name.
- Impersonates Amazon, a trusted company, to gain the user's trust.
- Creates a **false sense of urgency** by claiming the account will be suspended within 24 hours.
- Includes a **suspicious link** asking the user to update payment information.
- Requests **sensitive information**, such as payment details, through an email link.
- Uses **fear tactics** to pressure the user into acting without verifying the request.
- Encourages the user to click a button or link instead of visiting the official Amazon website directly.

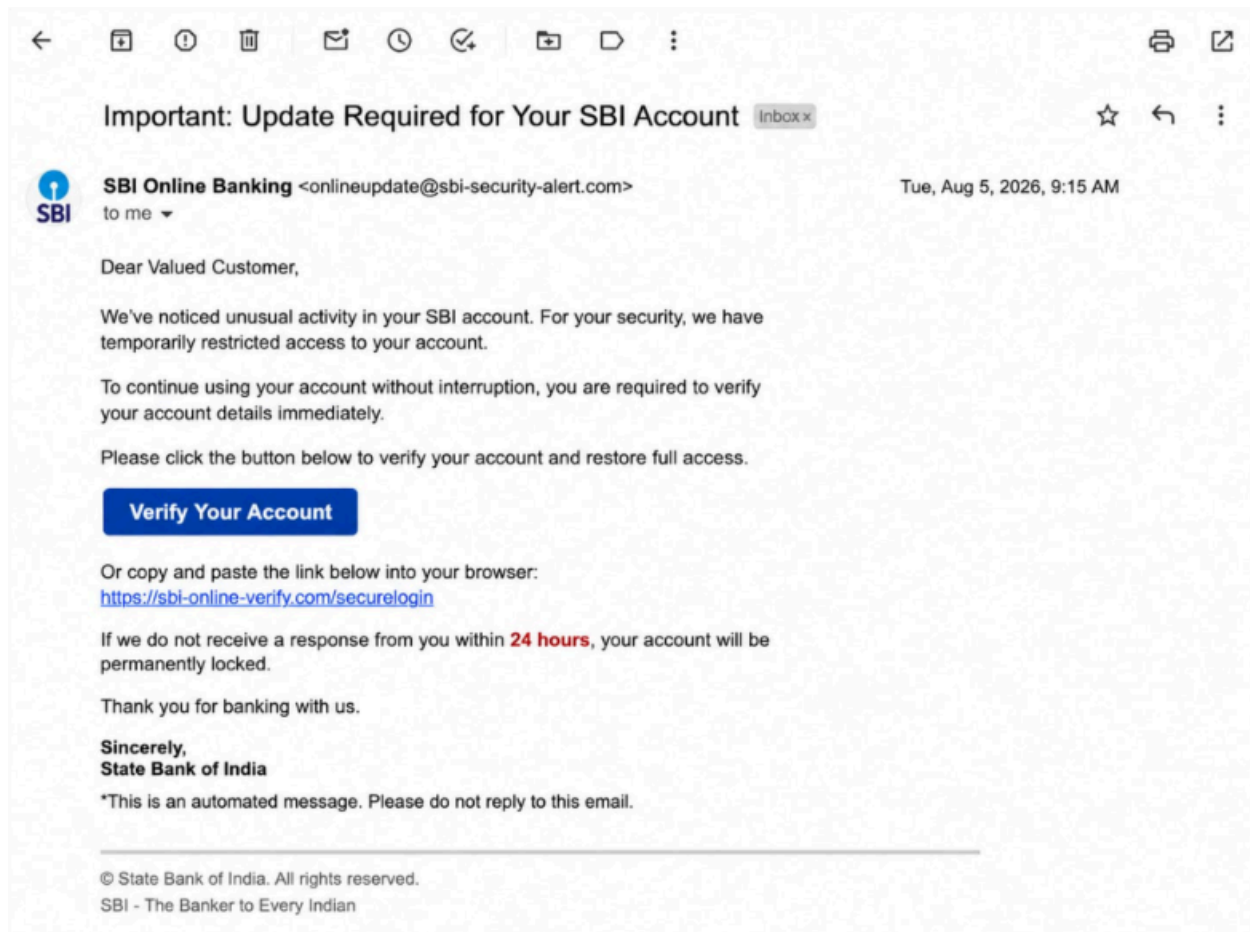
RECOMMENDED ACTIONS

- Do not click the "Update Payment Information" button or any links in the email.
- Do not enter your Amazon login credentials or payment details through links received by email.
- Visit Amazon directly by typing www.amazon.com into your browser or using the official Amazon app.
- Check your account for any genuine notifications through the official Amazon website.
- Report the email as phishing to Amazon or your organization's IT/security team.
- Delete the email after reporting it.
- Enable Multi-Factor Authentication (MFA) on your Amazon account for additional security.

CONCLUSION

This email is classified as a phishing email because it impersonates Amazon, creates a false sense of urgency, and attempts to trick users into revealing sensitive payment information through a suspicious link. Users should always verify such requests through the official Amazon website instead of clicking links in emails.

EMAIL SAMPLE 3 ANALYSIS



FIELD	DETAILS
Email Type	Phishing Email
Organization Impersonated	State Bank Of India
Target	SBI Customers
Risk Level	High
Classification	Phishing

PHISHING INDICATORS

- Uses a generic greeting ("Dear Valued Customer").
- Claims unusual account activity to create panic.
- Pressures the user to act within 24 hours.
- Includes a fake verification button and suspicious link.
- Requests account verification through an email link instead of the official banking website.
- Attempts to steal sensitive banking information by impersonating SBI.

RECOMMENDED ACTIONS

- Do not click the "**Verify Your Account**" button or any links in the email.
- Never enter banking credentials through links received by email.
- Visit the **official SBI website** or use the official SBI mobile app to verify your account status.
- Report the email to your bank or organization's IT/security team.
- Delete the email after reporting it.
- Enable **Multi-Factor Authentication (MFA)** and regularly monitor your bank account for suspicious activity.

CONCLUSION

This email is classified as Phishing because it impersonates a trusted bank, creates urgency, and attempts to trick users into revealing sensitive banking information through a fraudulent verification link.

PHISHING PREVENTION GUIDELINES

- Verify the sender's email address before responding.
- Avoid clicking on suspicious links or downloading unknown attachments.
- Hover over links to check the actual destination before clicking.
- Never share passwords, OTPs, or banking details through email.
- Enable Multi-Factor Authentication (MFA) on important accounts.
- Keep your operating system, browser, and antivirus software updated.
- Visit websites by typing the official URL instead of using email links.
- Report suspicious emails to your email provider or your organization's IT/security team.
- Delete phishing emails after reporting them.

CONCLUSION

This report analyzed three phishing email samples and identified common phishing indicators such as impersonation of trusted organizations, urgent language, suspicious links, and requests for sensitive information. The analysis demonstrated how phishing attacks attempt to deceive users into revealing confidential information.

By following recommended cybersecurity practices, verifying email authenticity, and staying alert to phishing techniques, users can significantly reduce the risk of falling victim to phishing attacks. Continuous awareness and user education are essential for maintaining a secure digital environment.

REFERENCES

1. OWASP Foundation – OWASP Security Resources
2. Microsoft Security Documentation
3. Amazon Security Help
4. State Bank of India Cyber Security Awareness
5. Future Interns – Cyber Security Internship Task Guidelines.

